

Exercise Sheet 4
 Lecture “Cybersecurity”
 Summer Semester 2026
 Practical Exercise: Asymmetric Cryptography

AES and RSA

In the lecture, you learned about the symmetric encryption method **AES** and the asymmetric method **RSA**. The basic procedure for **RSA** is summarized below.

Key Generation in 5 Steps:

1. Choose two large prime numbers p and q .
2. Compute the product N of both numbers: $N = p \cdot q$.
3. Compute Euler’s totient: $T = (p - 1) \cdot (q - 1)$.
4. Choose two numbers e and d , such that: $(e \cdot d) \bmod T = 1$
5. Public Key: (N, e) , and Private Key: (N, d)

Notes

- The number e must be smaller than the totient T : $e < T$
- The number e must be *coprime* (i.e. relatively prime) to both T and N .
- Make sure that for d : $(e \cdot d) \bmod T = 1$
- For a number/letter x (plaintext), it must hold that $x < N$

Encryption and Decryption with RSA

The value x is encrypted as:

$$y = x^e \bmod N$$

The value y is decrypted as:

$$x = y^d \bmod N$$

A	0	F	5	K	10	P	15	U	20	
B	1	G	6	L	11	Q	16	V	21	
C	2	H	7	M	12	R	17	W	22	Z 25
D	3	I	8	N	13	S	18	X	23	
E	4	J	9	O	14	T	19	Y	24	

Table 1: Caesar Alphabet, assigned to numbers (0-25)

Exercise 1.

- a) Given is an RSA encryption with: $p = 2$, $q = 7$, $N = 14$, $T = 6$, $e = 5$, and $d = 11$. Which pairs of numbers form the private key and which form the public key?
- b) Use the values from **a)** and the letter mapping from Table 1 to encrypt the message 'BCD' using RSA. Verify by decrypting the ciphertext and comparing it to the plaintext.
- c) Compare symmetric and asymmetric cryptography. What are the advantages and disadvantages of using symmetric and asymmetric cryptography? (A general comparison is sufficient, but you can also argue with specific examples such as symmetric encryption (*e.g.*, *AES*) and asymmetric encryption (*e.g.*, *RSA*).)
- d) A medium-sized company has 120 employees. How many keys are needed so that all employees can communicate directly (without additional infrastructure) with each other? Consider the following cases:
 - Only *AES* is used.
 - Only *RSA* is used.
- e) An encryption library such as *OpenSSL* achieves a performance of 100 kbit s^{-1} for **RSA** on your computer. For **AES** encryption, your computer achieves a performance of 17 Mbit s^{-1} . How long does it take to decrypt a video of size 1 GB? Consider each case for AES and RSA.
- f) Where does the performance difference between AES and RSA come from (see previous question)? What could be done to combine the advantages of asymmetric cryptography with those of symmetric cryptography?

Practical Exercise GPG (Bonus)

Encryption is essential for daily and secure communication, e.g., with your bank or one of the numerous online shops. In this exercise, you will learn how to encrypt arbitrary files using RSA in practice and exchange them with your friends.

Exercise 2.

- a) Install GPG¹ on your system. If you are using Windows, we recommend: *GPG4Win*², which is available free of charge.
- b) Create an RSA key pair (public and private) with a key length of at least 2048 bits. Detailed instructions for this can be found in the *GPG4Win* documentation: <https://files.gpg4win.org/doc/gpg4win-compendium-de.pdf>.
- c) Find a partner with whom you would like to exchange secret messages in the future. Exchange your public RSA keys, e.g., via USB stick or email. Import the public key under your partner's first name in GPG.
- d) Encrypt a file (e.g., a text file/Notepad) with secret content and send it via the Internet (e.g., email) to your partner.
- e) Decrypt the received message.

¹<https://gnupg.org/>

²<https://www.gpg4win.de/version4-de.html>